

Introduction: the nature of economics and the scope of microeconomics

**ECON 101: Introduction to
Microeconomics**

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THE POWER OF ECONOMIC ANALYSIS

- The economic way of thinking is a framework to analyze solutions to economic problems, such as:
- How the prices of labor, capital, and land are set in the economy, and how these prices are used to allocate resources.
- How financial markets allocate capital to the economy.
- How government regulations affect market efficiency.



THE POWER OF ECONOMIC ANALYSIS

Economics also examines how:

- income is distributed among the population.
- government spending, taxes, and debt affect growth.
- firms compete in different market structures.
- consumers allocate their income among different goods and services.



THE POWER OF ECONOMIC ANALYSIS

Also,

- Examines the patterns of trade among nations, and analyses the impact of trade barriers.
- Looks at the growth in developing countries, and proposes ways to encourage the efficient use of resources.



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But also, economics is a way of thinking about all decisions, such as:

- Fertility.
- Schooling.
- Crime.
- Having a child.



Defining Economics

Economics studies how
people make decisions!



Defining Economics

Is the study of how societies use scarce resources to produce valuable commodities and distribute them among different people (Samuelson, 1948)

In other words,

- Is the study of how people allocate their limited resources to satisfy their unlimited wants.

More generally,

- The study of how people make choices when there is scarcity.



Defining Economics

Resources

Things used to produce other things to satisfy people's wants.

Wants

What people like → represented by an **utility function**.



Utility Function

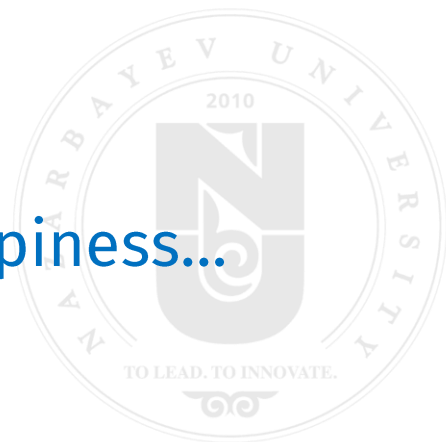
$$U = U(x_1, x_2, x_3, \dots)$$

X1: cars

X2: number of friends

X3: Going out on weekends,
and so on...

You can think about it as a measure of happiness...



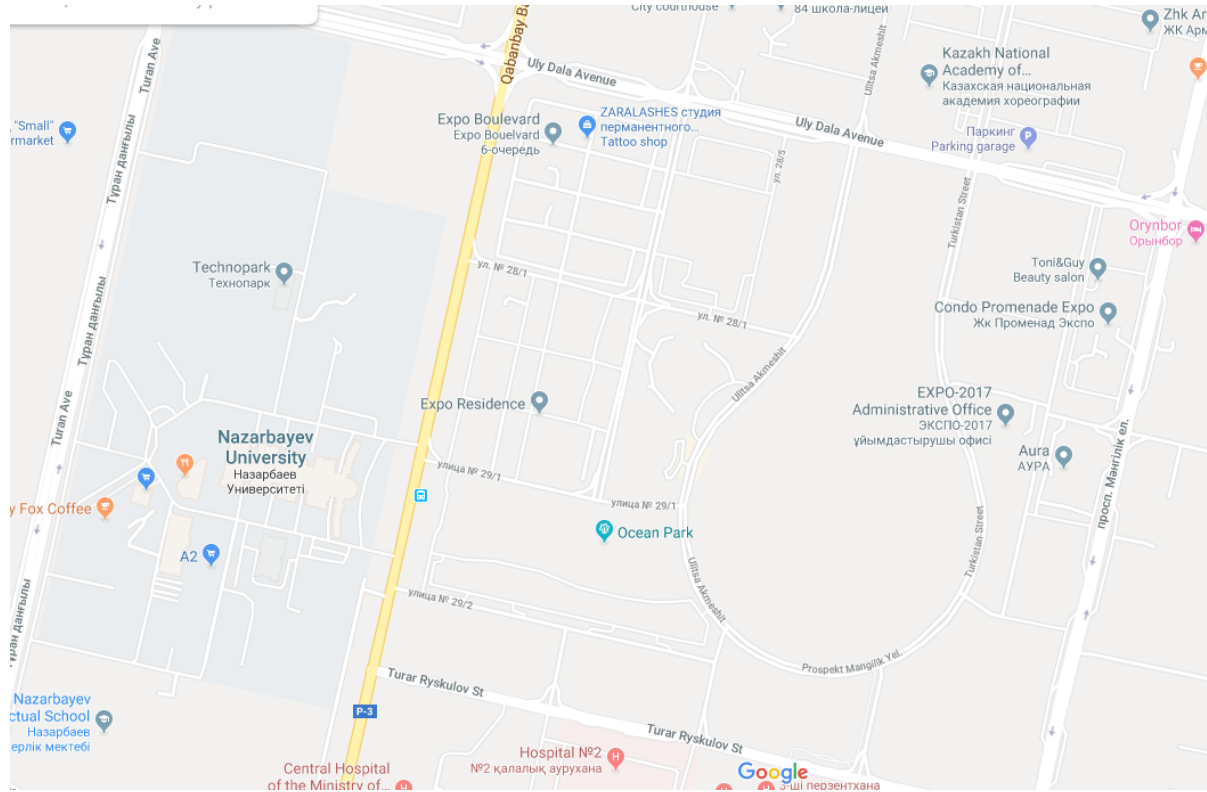
Economic Modeling

Economic Models

Models are simplified representations of the real world used as the basis for predictions or explanations



Example of a model



The purpose matters for model selection...



Economic Modeling

Assumptions

The set of circumstances in which
a model is applicable



Economic Modeling

Four important assumptions:

- Self-interest behavior
- Rationality
- Equilibrium
- Ceteris Paribus

Optimizing behavior
of decision-makers



Self-interest assumption

“It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from the regard to their own interest”

Adam Smith, An Inquiry into the Nature and Causes of the Wealth of Nations, 1776.

- Economic agents maximize their “utility” or “well-being.”



Self-interest assumption

This just means that we assume individuals follow the pursuit of their goals

- Humanitarian
- Prestige
- Wealth
- Power
- Friendship



Rationality Assumption

- People do not intentionally make decisions that would leave them worse off



The economic agent: rational self-interest

- The self-interest and the rationality assumptions imply that economic agents make decisions to maximize their utility subject to their constraints.
- In other words: economic agents are optimizers, i.e., they make the best possible choice according to their preferences and tastes with their given information and constraints.
- Economic agents evaluate costs and benefits



Example: is Economics good for you?

Costs

- Tuition (not if you have a scholarship)
- Time
- Other courses
- Sleep?
- Stress?

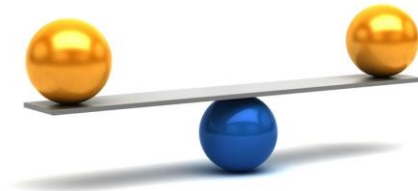
Benefits

- Graduation
- Knowledge
- Higher earnings potential?
- Learning to think like an Economist!



Equilibrium assumption

- In every economic model, there is a notion of **equilibrium**.
- A situation in which each economic agent, after choosing his/her optimal choices, does not benefit by changing his /her behaviour.



Ceteris Paribus assumption

- Nothing changes except the factor or factors being studied.
- A good model allows us to perform the “thought experiment” of what would happen if we change one of the factors but not the others!



Correlation vs Causation

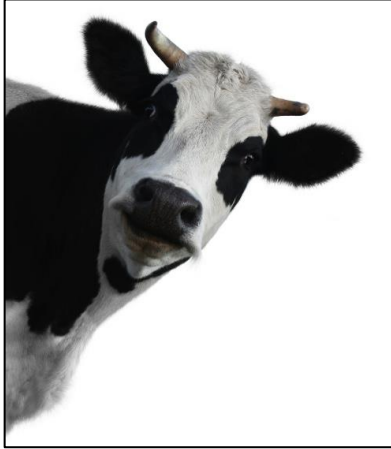
- The ceteris paribus thought experiment that a good model permits allows us to distinguish correlation vs causation.
- What is the difference between causation and correlation?



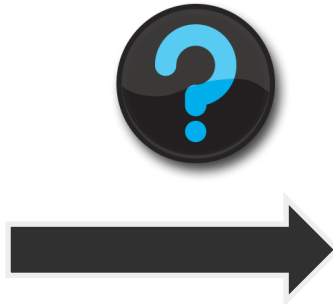
Correlation vs Causation

A study from England showed that cows who were named gave more milk than unnamed cows

Causation? or Correlation?



Buttercup the cow



Causation vs. Correlation

- **Causation** - When one thing directly affects another.
 - Example: pulling an all-nighter will make you tired
- **Correlation** - When two things are related
 - Positive correlation – they both change in the same direction
 - Negative correlation – they change in opposite directions



Causation vs Correlation

Why isn't correlation the same thing as causality?

1. Omitted variables

If we ignore something that contributes to cause and effect, then that something is an omitted variable. A correlation might not make sense until the omitted variable is added.

2. Reverse causality

Reverse causality is when there is cause and effect, but it goes in the opposite direction as what we thought.

A good model allows us to discern if we have these issues!



Economics as a Science/Economics as an Art

- Economics is an **empirical** science
 - Real-world data is used to evaluate the usefulness of a model.
 - Models are useful if they predict economic phenomena.
- Constructing a “good” model takes talent and experience, and is more akin to **art**.
- Economic models predict how people react, not how they think.



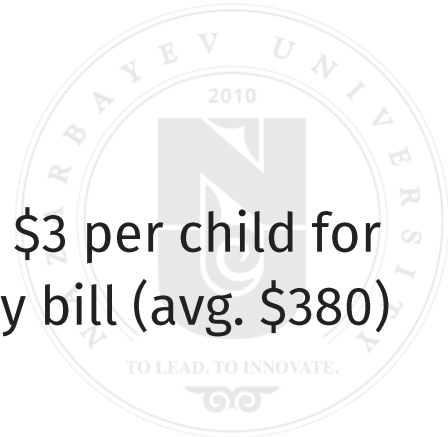
A story: a day-care in Haifa

The problem:

- Children are supposed to be picked up by 4:00 p.m.
- Some parents are late → Anxious children
- At least one teacher that has to wait until all parents arrive

The proposal:

- Two Economists suggested to fine tardy parents
- Any parent arriving more than 10 min. late would pay \$3 per child for each incident. The fee would be added to the monthly bill (avg. \$380)



A story: a day-care in Haifa

The result:

- Before the fine: Avg. 8 late pickups per week per day-care center
- After the fine: Avg. 20 late pickups per week....!!! Why???

Incentives!!!



Incentives

- Individuals respond to incentives by weighting costs and benefits. An incentive is a mean of inducing people to do more of some action and less of a another one.
- We must be careful about the behavior that incentives generate!



Incentives

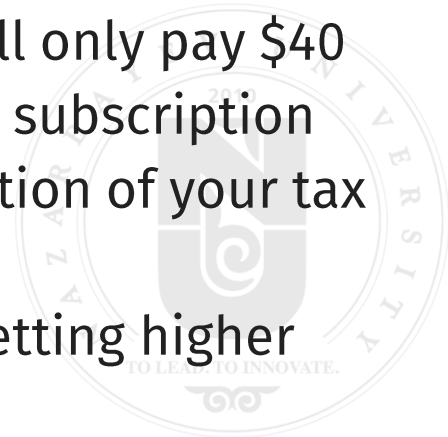
- **Positive incentives** encourage an action by rewarding some behavior (more of a good thing).
- **Negative incentives** encourage an action by penalizing some behavior (less of a bad thing).



Incentives

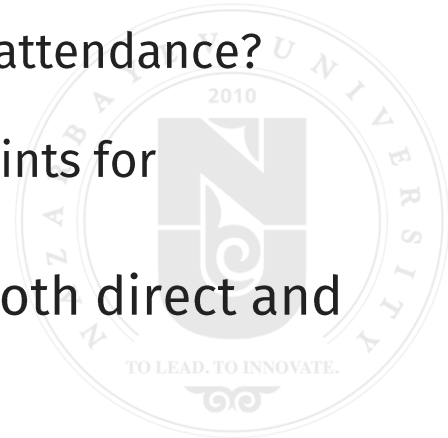
Some examples of incentives

- A relative will buy you a laptop you get an A in ECON 101
- A relative will take your car away if you get a D in ECON 101
- Discounts: if you sign in for 6 months at the gym you'll only pay \$40 per month instead of paying \$60 if you get a monthly subscription
- Tax amnesties: the government will forgive some portion of your tax duties if you pay before certain date.
- The KZ govt. will impose higher tariffs on countries setting higher tariffs on Kazakhstani imports.



Incentives

- Sometimes economists refer to **direct incentives** and **indirect incentives** to refer to intended consequences and unintended consequences on behavior.
- Question for you:
 - What is the direct incentive of giving bonus points for attendance?
 - What could be an indirect incentive of giving bonus points for attendance?
- Good economic policy making requires considering both direct and indirect incentives (As the daycare story told us).



Microeconomics vs Macroeconomics

Microeconomics

- The study of decision making undertaken by individuals (or households) and by firms.
- Study of behavior of economic agents at the individual level.
 - Investment in R&D
 - Crime
 - Education
 - Business Strategy.

Macroeconomics

- The study of the behavior of the economy as a whole
 - Inflation
 - Business Cycles – recessions, peaks, etc.
 - Taxes
 - Unemployment



Positive vs Normative Economic analysis

Positive Economics

- Purely descriptive statements or scientific predictions—a statement of what is

Normative Economics

- Analysis involving value judgments—a statement of what ought to be





End of Lecture 1.
Read Chapter 1 from the book!